

**BACHELOR OF COMPUTER
APPLICATIONS (BCA)
(Revised)**

Term-End Examination

June, 2023

**BCS-042 : INTRODUCTION TO
ALGORITHM DESIGN**

Time : 2 Hours

Maximum Marks : 50

Note : (i) *Question No. 1 is compulsory.*

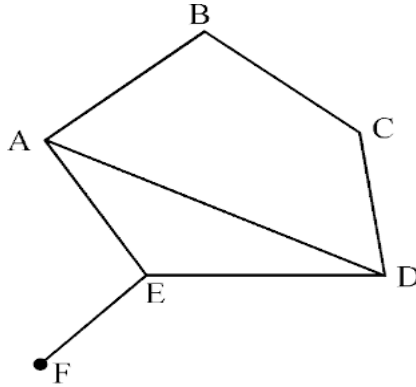
(ii) *Answer any **three** questions from the rest.*

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1. (a) What is complexity of algorithm ? Explain space complexity and time complexity of algorithms with the help of example. 5
 - (b) Write linear search algorithm and do analysis of this algorithm for best case and worst case. 5

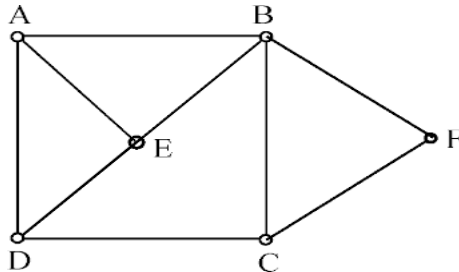
- (c) Using mathematical induction method, show that for all positive integers n :

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- (d) What is Adjacency matrix ? Write adjacency matrix for the following graph :



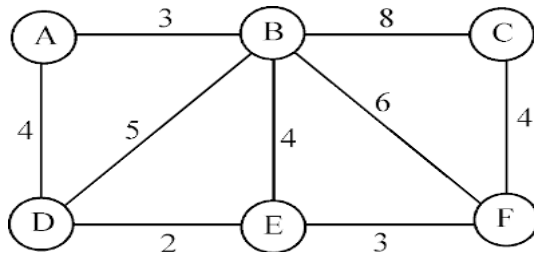
2. (a) Write Depth-First Search (DFS) algorithm. Also traverse the following graph using DFS from node A.



- (b) Solve the following recurrence relation using recurrence tree method : 3

$$T(n) = 4T\left(\frac{n}{2}\right) + n$$

3. (a) Write Kruskal's algorithm for finding Minimum Cost Spanning Tree (MCST). Find MCST of the following graph using Kruskal's algorithm. 8



- (b) Explain use of Big oh (O) notation in the analysis of algorithms with example. 2
4. (a) Find the optimal solution to the knapsack (fractional) problem for $n = 5$ and $m = 10$, where n is the number of objects and m is the capacity of the knapsack.

Profit and weight of each object are given below : 6

$$(P_1, P_2, P_3, P_4, P_5) = (10, 30, 35, 20, 40)$$

$$(W_1, W_2, W_3, W_4, W_5) = (3, 5, 2, 6, 11)$$

- (b) In context of algorithm study, explain the following with the help of an example of each : 4

(i) Upper Bound

(ii) Backtracking

5. (a) Sort the following list using Bubble sort algorithm. Show the steps of sorting : 6

30, 8, 7, 14, 20, 28, 10, 6

- (b) Write algorithm for adding two matrices of order $m \times n$ and find its time complexity. 4

No. of Printed Pages : 3

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December, 2023

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) Show that : 2

$$5n^2 + 2n + 2 = \theta(n^2)$$

(b) Solve the following recurrence relation : 3

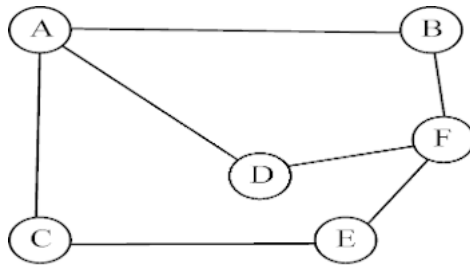
$$T(n) = 3T\left(\frac{n}{2}\right) + n$$

(c) Write binary search algorithm and explain its best case and worst case complexity. 5

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[2]**BCS-042**

- (d) Represent the following graph using Adjacency List : 4



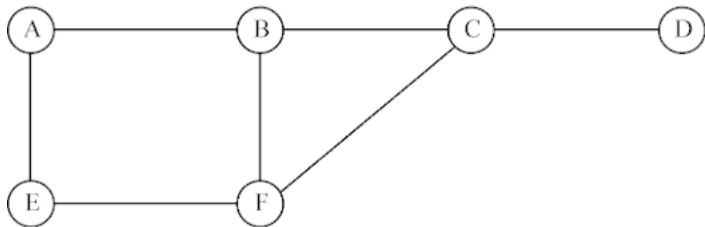
- (e) Sort the following list of elements using quick sort. Also show the steps of the operation : 6

8, 29, 30, 6, 25, 7, 8, 9

2. (a) Define the following terms : 4
- (i) Connected graph
 - (ii) Cycle in an undirected graph
- (b) Write Euclid's algorithm for finding GCD and explain it. 6
3. (a) What is spanning tree ? Explain application of spanning tree. 4
- (b) Write and explain Dijkstra's single source shortest path algorithm. 6

[3]

4. (a) Briefly explain Greedy technique for designing of algorithm. 4
- (b) Write DFS traversal sequence from the node A for the graph : 6



5. (a) Using mathematical induction method, show that for all positive integers n : 5

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

- (b) Write Prim's algorithm for finding minimum spanning tree and find the complexity of this algorithm. 5

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Note : (i) *Question No. 1 is compulsory and carries 20 marks.*

(ii) *Answer any **three** questions from the rest.*

1. (a) Arrange the following time complexities in the increasing order : 2

$\log n, O(n^2), O(3^n), n!$

- (b) Write the names of the following symbols : 3

(i) Ω

(ii) θ

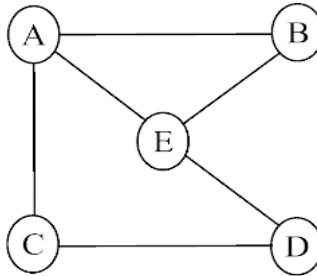
(iii) O

- (c) What is recurrence relation ? Define Fibonacci sequence using recurrence relation. 4
- (d) What is linear search method ? Apply linear search to search the number 9 in the following list of numbers. Show the searching steps : 6

8, 4, 10, 6, 9, 2

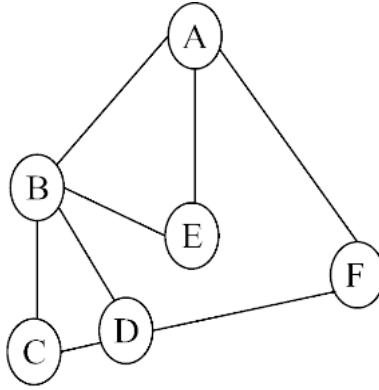
Also, analyse the worst case complexity of the linear search method.

- (e) Write adjacency-list and adjacency-matrix representation of the following graph : 5



2. (a) Explain the following terms : 3
- (i) Space complexity
 - (ii) Cycle in a graph
 - (iii) Lower bound

- (b) Traverse the following graph using DFS-method, taking A as a starting vertex and sequence of vertices in the order of their appearance in traversal : 7



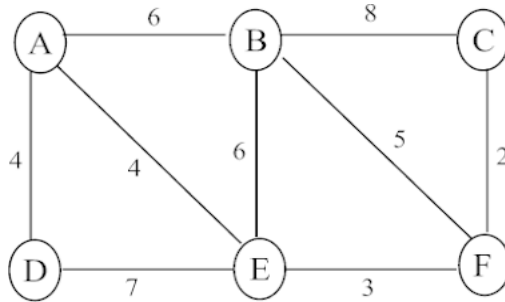
Also, write the time complexity of DFS method.

3. (a) Write and explain binary search algorithm. Explain its time complexity for best and worst cases. 5
- (b) Prove the following propositions using mathematical induction : 5

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

4. (a) Define the term algorithm. Explain various properties of an algorithm. 4

- (b) Write an algorithm for multiplying two matrices. Also find the time complexity of this algorithm. 6
5. (a) Write Kruskal's algorithm for finding minimum cost spanning tree (MCST). Apply this algorithm on the following graph to find MCST : 8



- (b) Briefly explain divide and conquer approach of problem solving. 2

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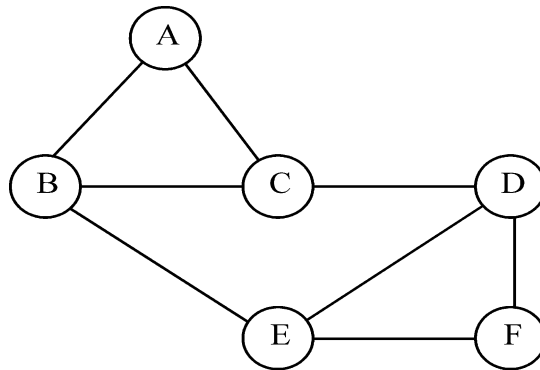
(ii) *Answer any **three** questions from the rest.*

1. (a) Define O (big oh) notation and prove or disprove the following using the basic definition of O (big oh) : 4

$$5n^3 + 2n^2 + 8 = O(n^3)$$

- (b) Write bubble sort algorithm and find its worst case time complexity. 5

- (c) What is connected-graph ? Explain with the help of an example. 2
- (d) Write adjacency list and adjacency matrix representation of the following graph : 5



- (e) Solve the following recurrence relation, using substitution method : 4

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

2. (a) Define the following terms : 2
- (i) Tree
 - (ii) Directed Graph
- (b) Write a recurrence relation for the recursive factorial function. 3
- (c) Explain the use of lower bound and upper bound concepts with the help of example. 5

3. (a) Write Euclid's GCD algorithm. Also, find GCD (592, 252) using Euclid's algorithm. 5
- (b) Sort the following list using selection sort. Also, show all the intermediate steps in the process : 5

2, 10, 6, 18, 25, 9

4. (a) Define a fractional knapsack problem. Find the optimal solution to the following instance of a knapsack problem and show the stepwise running of the algorithm : 8

No. of objects = 5.

Capacity of a knapsack : $M = 18$
 $(P_1, P_2, \dots, P_7) = (12, 10, 8, 11, 14, 7, 9)$

where P_i is a profit and

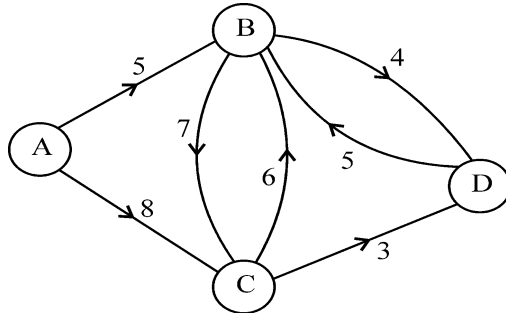
$(W_1, W_2, \dots, W_7) = (4, 6, 5, 7, 2, 3, 6),$

where W_i is a weight.

Each object has a profit P_i and weight W_i .

- (b) What is binary search ? 2

5. Apply Dijkstra's algorithm to find the shortest path from source vertex A to each of the other vertices of the following directed graph : 10



Show all the steps of the solution.

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Note : *Question No. 1 is compulsory and carries
20 marks. Answer any **three** questions
from the rest.*

1. (a) Define the term 'algorithm' and explain its importance in Computer Science. Discuss the concept of recursion and provide an example of recursive algorithm. 5

(b) What is the time complexity of linear search algorithm ? Explain with the help of an example. 5

(c) Describe any *three* of the following :

$$3 \times 2 = 6$$

(i) Backtracking

(ii) Fractional Knapsack problem

(iii) Connected graph

(iv) Euclid Algorithm for GCD

- (d) Suppose you are given currency notes of the following denominations :

(2, 5, 10, 15, 20, 100, 500)

Further it is assumed that currency notes of each denomination are available in sufficient numbers for purpose of using the minimum number of notes. The problem is to find minimum number of notes to make an amount of ₹ 437. Using greedy approach, show sequence of steps for selection of notes to make ₹ 437. 4

2. (a) Solve the recurrence : 5

$$T(n) = T\left(\left\lfloor \frac{n}{3} \right\rfloor\right) + T\left(\left\lfloor \frac{2n}{3} \right\rfloor\right) + n$$

using recursion tree method.

- (b) The recurrence $T(n) = 7T\left(\frac{n}{2}\right) + n^2$

describes the runtime of an algorithm.

A competing algorithm A' has running

time of $T'(n) = \alpha T\left(\frac{n}{4}\right) + n^2$. What is the

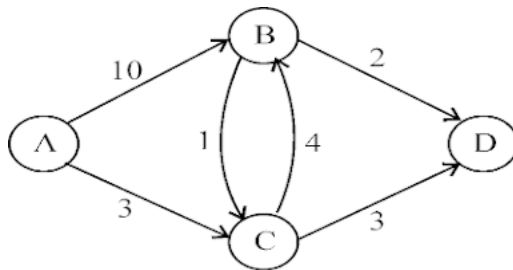
largest integer value for $|\alpha|$ such that

$|A'|$ is asymptotically faster than A ? 5

3. (a) Given an array of size(n), analyze the time complexity of the bubble sort algorithm. If array is already sorted in ascending order, what would be the best time complexity ? Provide the Big-O notation for both the best case and worst case scenarios. 5

- (b) Apply the Dijkstra's algorithm to find the shortest path from Vertex A to each of the other vertices for the following directed graph :

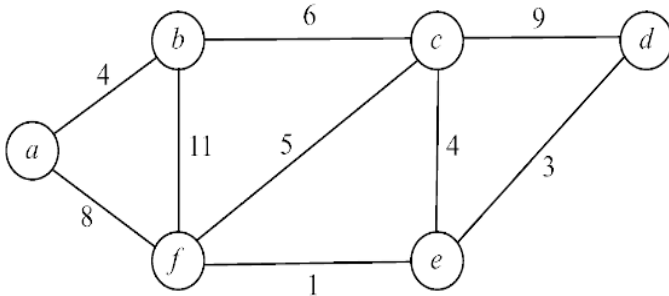
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4. (a) Compare and contrast Depth-First Search (DFS) and Breadth First Search (BFS) algorithms. Highlight their similarities and differences and discuss their respective use cases.
- (b) Is the minimum spanning tree of any graph unique ? Apply Prim's algorithm

5

to find a minimum cost spanning tree
for the following graph ('a' is the
starting vertex) : 5



5. (a) Show that the runtime of partition procedure of quick sort algorithm on a sub-array of size (n) is $\theta(n)$. 5
- (b) Explain the divide and conquer approach of matrix multiplication of two 4×4 matrices. 5

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No. of Printed Pages : 5

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1. (a) Explain the concept of the space complexity in context of algorithm analysis. How is it different from time complexity ? Provide an example to illustrate the calculation of space complexity. 5
- (b) Given a list of distinct integers, write an algorithm to determine the position of an integer in the list using linear search and count the number of comparison operations required. 5
- (c) Describe any *three* of the following :
3×2=6
 - (i) Greedy technique
 - (ii) Cycle in a Directed Graph
 - (iii) Upper Bound
 - (iv) Branch and Bound

- (d) Use Master Theorem to give tight asymptotic bounds of the following :

$$2 \times 2 = 4$$

(i) $T(n) = 4T\left(\frac{n}{2}\right) + n^2$

(ii) $T(n) = 2T\left(\frac{n}{2}\right) + n\sqrt{n}$

2. (a) State Euclid's algorithm and find GCD (595, 252) by this algorithm. 5

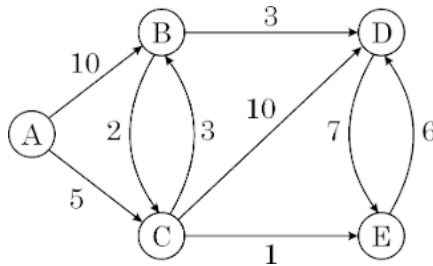
- (b) Write the iterations for sorting the following list of numbers using bubble sort and selection sort : 5

45, 67, 13, 90, 2, 35, 28, 10

3. (a) Differentiate between Kruskal's algorithm and Prim's algorithm. 4

- (b) Apply Dijkstra's algorithm to find shortest path from source vertex A to each of the other vertices of the following directed graph :

6



4. (a) Consider the following sorted array A with 13 elements :

7, 15, 19, 27, 30, 48, 56, 59, 75, 87, 94,
110, 111

Illustrate the working of Binary search algorithm while searching for ITEM :

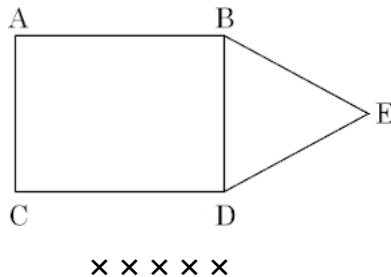
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(i) 19

(ii) 94

- (b) Show that the runtime of QUICK SORT algorithm in the best case is $\theta(n \log n)$. 5
5. (a) Multiply 1026732×0732920 using divide and conquer technique using Karatsuba method. 5
- (b) Write an algorithm for Depth First Search (DFS) and traverse the following graph using DFS. Starting vertex is A :

5



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BCS-042

**BACHELOR OF COMPUTER
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Term-End Examination

June, 2026

**BCS-042 : INTRODUCTION TO
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Time : 2 Hours

Maximum Marks : 50

Note : *Question No. 1 is compulsory and carries
20 marks. Answer any **three** questions
from the rest.*

1. (a) Prove the following proposition using induction : 5

$$P(n) : 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- (b) Solve the recurrence relation

$$T(n) = T\left(\left\lceil \frac{n}{3} \right\rceil\right) + T\left(\left\lfloor \frac{2n}{3} \right\rfloor\right) + n$$

using recursion tree method. 5

- (c) Why is analysis of an algorithm important ? Justify your answer with example. 5

- (d) Explain asymptotic notations. Also, discuss what is upper and lower bound in asymptotic notations. 5

2. (a) Write an algorithm for bubble sort and write its worst case and best case analysis. 6

(b) Find $\text{GCD}(595, 252)$ by Euclid's algorithm. 4

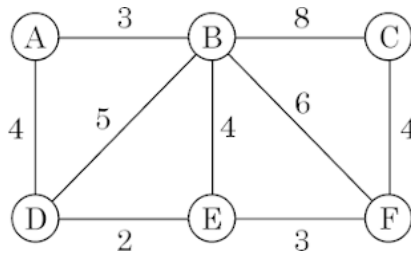
3. (a) Write basic algorithm to evaluate a polynomial and find its complexity. 5

(b) Find the optimal solution for the Knapsack instance $n = 7$ and $m = 15$: 5

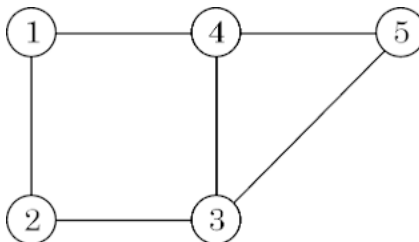
$$(P_1, P_2, P_3, \dots, P_7) = (10, 5, 15, 7, 6, 18, 3)$$

$$(W_1, W_2, W_3, \dots, W_7) = (2, 3, 5, 7, 1, 4, 1)$$

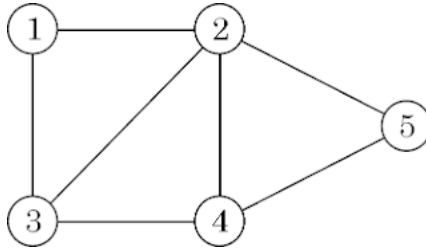
4. (a) Apply Kruskal's and Prim's algorithm to find a minimum cost of a spanning tree of the graph given below : 8



- (b) Define divide and conquer approach. 2
5. (a) What is Adjacency Matrix ? Write adjacency matrix and adjacency list for the following graph. 5



- (b) Write Depth-First Search (DFS) algorithm. Also traverse the following graph using DFS from node 1. 5



x x x x x